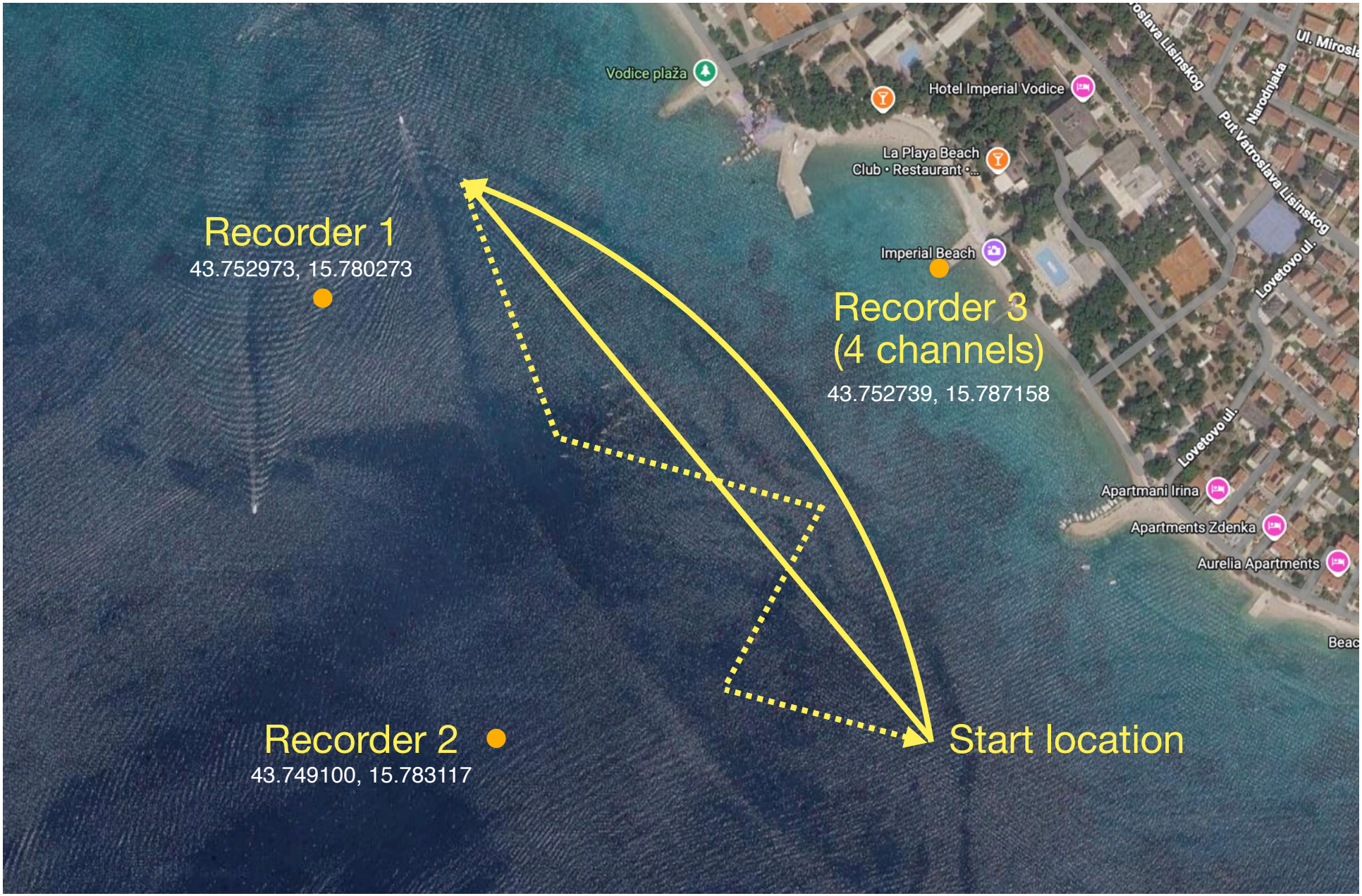


BTS CHALLENGE

ACOUSTIC TRACKING AND COURSE ESTIMATION OF A MOTORBOAT


Recorder 1
43.752973, 15.780273
Recorder 3
(4 channels)
43.752739, 15.787158
Recorder 2
43.749100, 15.783117
Start location
Vodice plaža
Hotel Imperial Vodice
La Playa Beach
Club • Restaurant •...
Imperial Beach
Apartmani Irina
Apartments Zdenka
Aurelia Apartments
Lovetovo ul.
Put Vatroslava Lisinskog
Narodnjaka

Task

Estimation motorboat path from acoustic recordings

- 3 motorboat runs:
  - Straight line run with pinger
  - Zig-zag run with pinger
  - Curved run without pinger (bonus)
- You will be given:
  - Acoustic recordings from the 3 recording systems
  - Start time and duration for the 3 runs

Submission

Deadline: Thursday, 22 October 2026, 6pm

- **Primary submission**
  - CSV file containing (time, latitude, longitude) of straight run with pinger
  - CSV file containing (time, latitude, longitude) of zig-zag run with pinger
- **Bonus submission**
  - CSV file containing (time, latitude, longitude) of run without pinger

Evaluation Criteria

1. **Tracking Accuracy** (50%): Calculated using Root Mean Square Error (RMSE) against the ground-truth GPS log for all three tracks
2. **Algorithmic Design** (30%): Quality of your chosen pinger waveform, multi-sensor fusion (LBL + USBL), and passive tracking approach
3. **Presentation** (20%): Clarity, presentation material, description of engineering decisions

Presentation Expectations (5—10min)

1. **Signal Design:** Explain your choice of pinger waveform(s), frequency range, etc.
2. **Tracking Strategy:** Show how you chose/fused LBL ranges and USBL angles (e.g., processing pipeline, your choice of filters and algorithms, …)
3. **Passive Tracking:** Explain your method for tracking the boat using only its own engine/propeller noise
4. **Visuals:** Focus on flowcharts, block diagrams, and your final estimated tracks—do not show raw code